

Test 7: fourth-order response

Test 7 fourth-order follow-up

Run run-29c4c4228ffddb4; analysis analysis-0001-ac36b679; classification pass.
Unchanged SS OCF 1 action and Test 6 clock calibration. Original Test 7 remains failed.
Prior Test 7 comparison is an inspected-data diagnostic; held-out wave is the prospective check.
No two-object recoil, coordinate invariance, gravity, angular stability, or emergent spacetime is tested.

Held-out interval: actual -2.087776737e-07; order-two -2.093903858e-07; order-four -2.089527184e-07 cycles.
Numerical interval budget 1.295e-09 cycles.

conservation: pass
heldout-interval: pass
heldout-waveform: pass
marker-timing: pass
prediction-first: pass
record-resolution: pass
sign-even: pass

prior-diagnostic

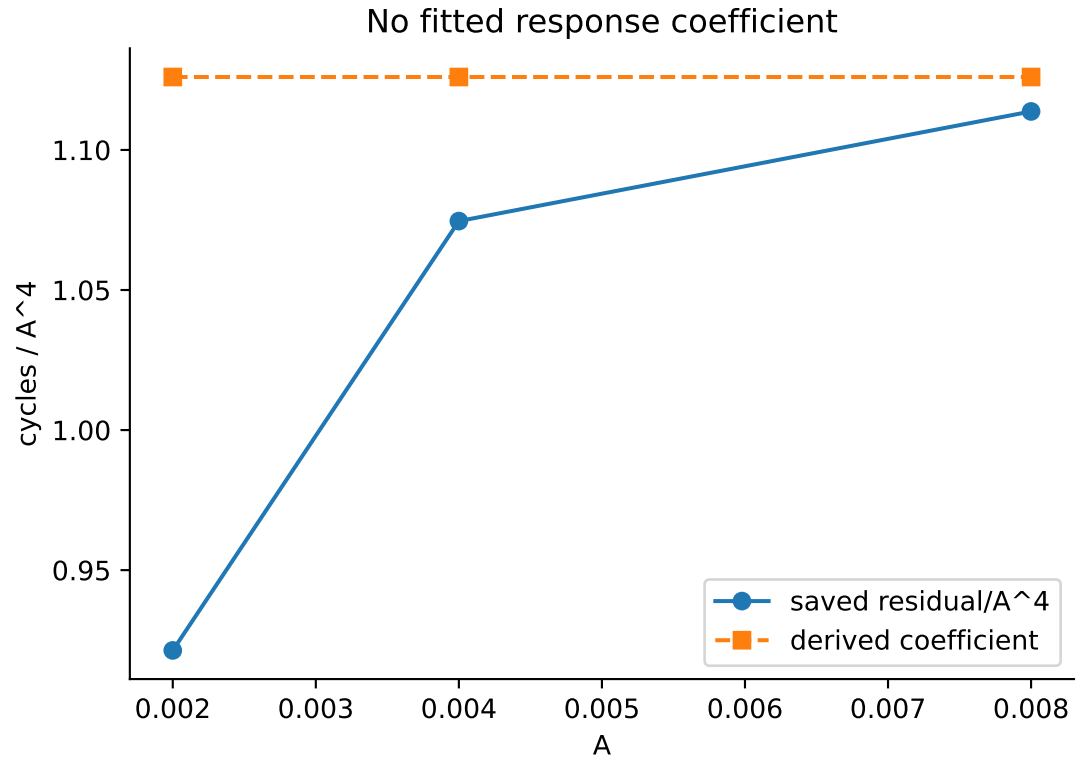
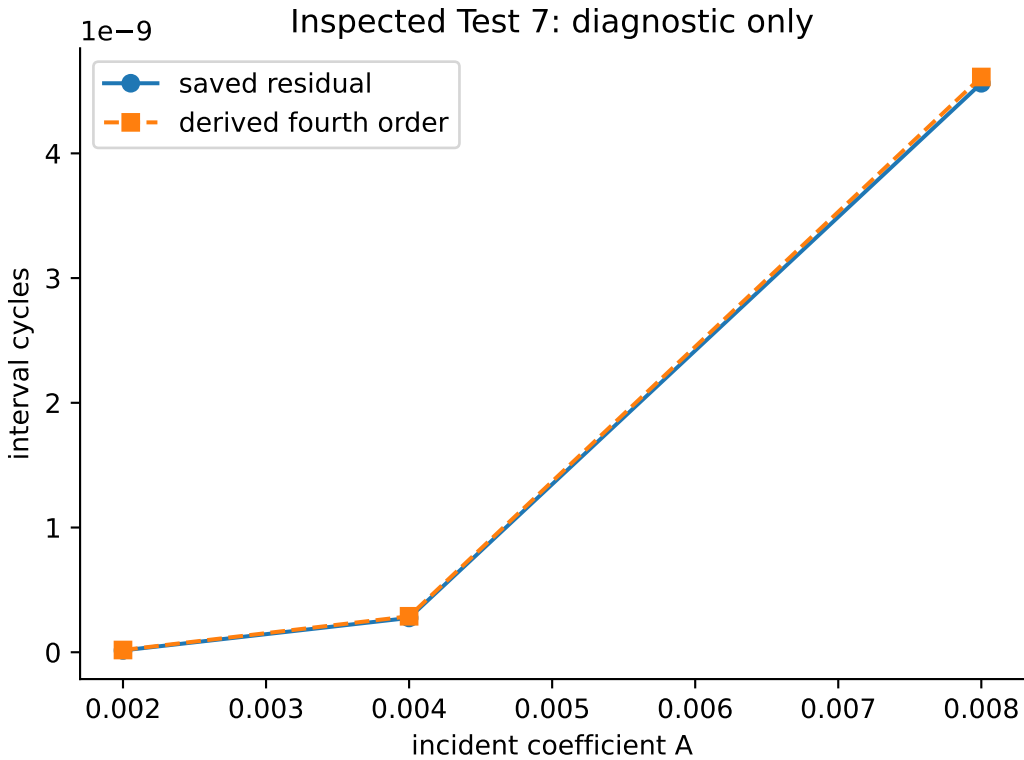
Question: Does the derived fourth-order coefficient explain the inspected residual sign and scaling?
Reading: Compare the saved Test 7 amplitude ladder at the same nominal $A=.004$ marker times.
Significance: This diagnoses the missing order independently of a fitted coefficient.
Limitation: The prior histories and marker times were already inspected; this is not a prospective prediction.

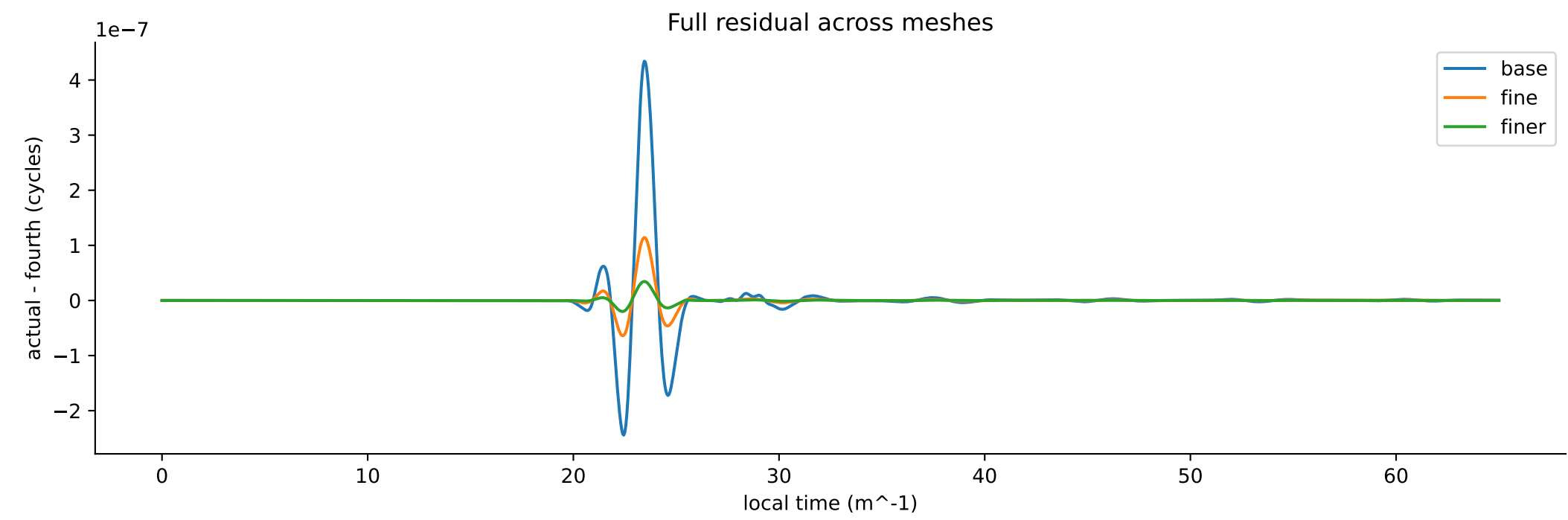
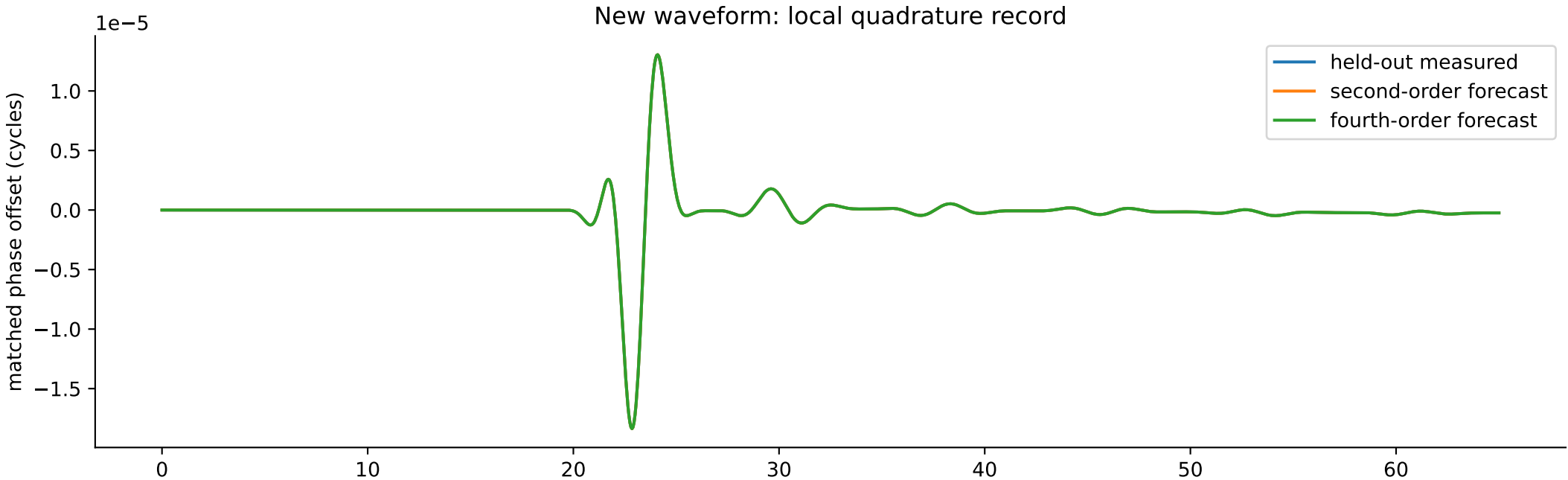
heldout-phase

Question: Does the locked fourth-order prediction improve a new local clock history?
Reading: Held-out finest trace error: second 0.1882%, fourth 0.1712%.
Significance: A separate waveform tests the coefficient after the inspected-history diagnostic.
Limitation: Quiet phase at the pulse markers is counterfactual; grid differences and inversion error remain.

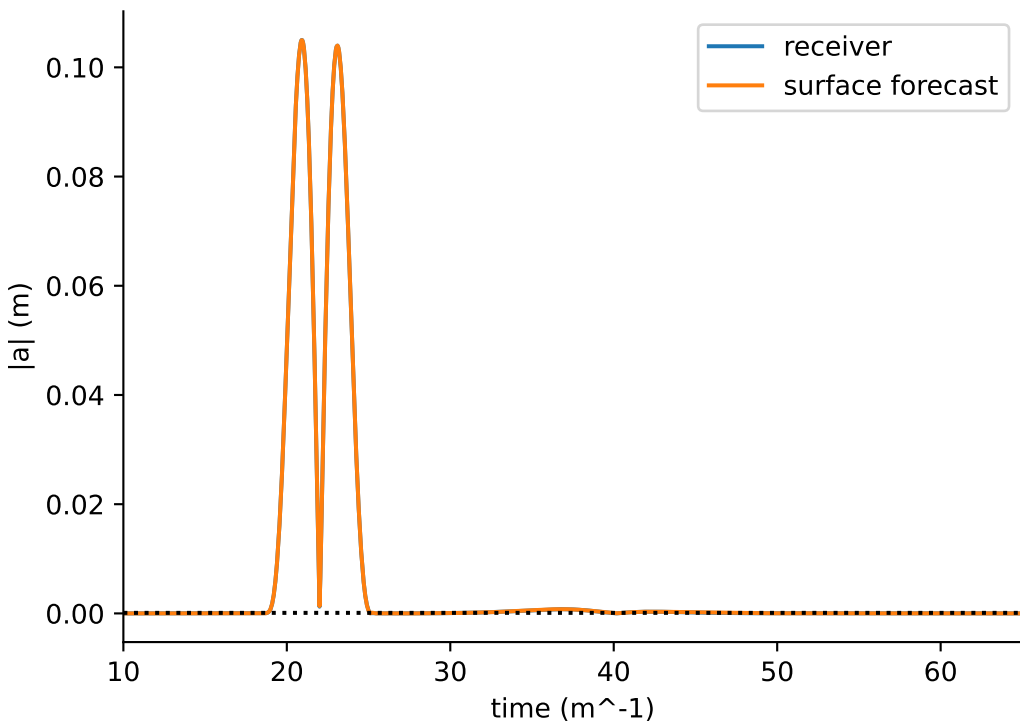
marker-reconstruction

Question: Do derivative-informed markers and twice-frequent output explain local timing error?
Reading: On $h=.025$, compare markers at 0.05 and decimated 0.1; coarse vacuum/known-Z maximum errors are 0.05998/0.05981.
Significance: Separates temporal interpolation from surface-propagation error.
Limitation: The coarse vacuum control is post-hoc; Hermite cannot repair spatial dispersion or WKB error.





Local threshold events

Magnified timing residual on $h=0.025$ 